📊 Spatio-Temporal IPL Analysis (2008–Present)

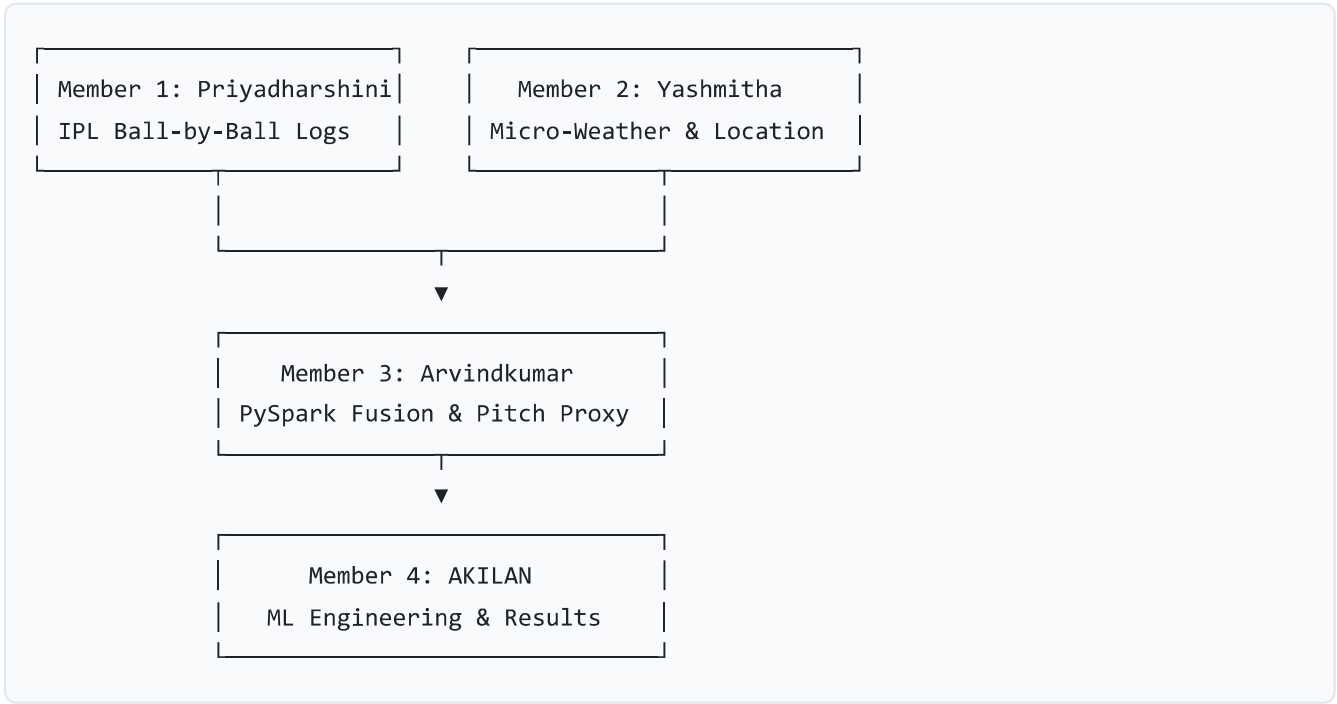
Comprehensive ML & Results Execution Report

Role: Member 4 – Machine Learning & Results (Akilan Pon Ilango)
Institution: SRM Institute of Science and Technology
Project: Spatio-Temporal Impact of Micro-Weather and Pitch Degradation on IPL Bowler Performance

📌 1. Executive Summary & Workflow Context

In this project, four modular roles collaborate to analyze and predict IPL bowling metrics:

1. **Priyadharshini (IPL Dataset):** Ball-by-ball extraction, player standardization, batting/bowling statistics.
2. **Yashmitha (Weather & Location):** Venue GPS mapping, micro-weather collection (temperature, humidity, dew, wind).
3. **Arvindkumar (PySpark & Data Fusion):** Spatial-temporal joining of match logs with weather timestamps and generating pitch degradation proxies.
4. **Akilan (You - ML & Results):** Ingests the fused dataset, designs the feature matrix, trains machine learning models, executes the season-based temporal evaluation (Train: 2008–2022, Test: 2023–2026), quantifies environmental impact, and produces all evaluation charts and reports.



🛠️ 2. Step-by-Step ML Roadmap From Scratch

🔹 STEP 1: Ingest & Audit the Integrated Dataset

- **Objective:** Load the unified dataset provided by the Data Fusion module (`data/fused_ipl_weather_pitch.csv` or equivalent).
- **Key Checks:**
    1. Confirm rows correspond to **Bowler Spells per Match** (or ball-by-ball aggregated to spells).
    2. Inspect missing value counts in weather columns (`temperature`, `humidity`, `dew_point`, `wind_speed`) and pitch wear metrics.
    3. Validate date ranges spanning **2008 through 2026**.

🔹 STEP 2: Feature Selection & Multimodal Matrix Design

You will construct a multi-modal feature vector containing three distinct signal sources:

A. Target Variables (Prediction Objectives)

1. `economy_rate`: Runs conceded per legal over ($6 \times \frac{\text{Runs Conceded}}{\text{Legal Balls}}$6 × Legal BallsRuns Conceded).

2. `boundary_frequency` : Number of 4s + 6s conceded per over ($6 \times \frac{\text{Boundaries Conceded}}{\text{Legal Balls}}$ 6 × Legal BallsBoundaries Conceded).

B. Independent Feature Groups

- **1. Cricket & Match Dynamics:**
  - `innings` (1st or 2nd innings — chasing pressure)
  - `powerplay_balls_ratio` (Overs 1–6)
  - `middle_balls_ratio` (Overs 7–15)
  - `death_balls_ratio` (Overs 16–20)
  - `bowler_career_economy` (Historical economy prior to match date — leakage-free)
  - `bowler_career_boundary_freq` (Historical boundary rate prior to match date)
  - `bowler_strike_rate` (Balls per wicket taken prior to match date)
  - `opponent_avg_runs_per_over` (Opponent batting form)
  - `venue_historical_avg_economy` (Base venue scoring difficulty)
- **2. Spatio-Temporal Weather Signals:**
  - `temperature` (°C)
  - `relative_humidity` (%)
  - `dew_point` (°C)
  - `dew_spread` ($\text{Temperature} - \text{Dew Point}$ Temperature − Dew Point — lower values indicate heavy dew likelihood)
  - `wind_speed` (km/h)
  - `is_night_match` (1 for evening games, 0 for day games)
- **3. Pitch Degradation Signals:**
  - `pitch_degradation_proxy` (Cumulative overs bowled on the specific pitch across the season or match)
  - `innings_pitch_wear` (Whether the pitch is fresh in innings 1 or deteriorated by innings 2)

🔹 STEP 3: Out-of-Time Temporal Train/Test Split

Standard random k-fold cross validation causes severe temporal data leakage in sports time-series. As specified in the project charter, you will perform a strict **Season-Wise Chronological Split**:

$\text{Training Set: Seasons } 2008 \rightarrow 2022 \quad (\approx 80\text{–}85\% \text{ of historical matches})$
Training Set: Seasons 2008 → 2022 (≈ 80–85% of historical matches)

Testing Set: Seasons 2023 → 2026 (Real-world out-of-time evaluation)
Testing Set: Seasons 2023 → 2026 (Real-world out-of-time evaluation)

🔹 STEP 4: Scikit-Learn Preprocessing Pipeline

To guarantee reproducibility and zero leakage, wrap preprocessing in a `ColumnTransformer`:

- **Numerical Features:** Imputed with median/mean using `SimpleImputer` + Normalized using `StandardScaler`.
- **Categorical Features (`bowling_team`, `opponent_team`, `venue`, `bowler_type`):** Encoded using `OneHotEncoder(handle_unknown='ignore')`.

🔹 STEP 5: Model Training & Benchmarking

Train a progression of regression models for both target variables:

1. **Linear Regression (Baseline):** Verifies linear relationships between environmental conditions and scoring rates.
2. **Random Forest Regressor:** Captures non-linear feature interactions (e.g., high humidity × × death overs).
3. **XGBoost / LightGBM Regressor:** Gradient-boosted decision trees for optimal tabular prediction accuracy.

🔹 STEP 6: Model Evaluation & Performance Metrics

Evaluate all models strictly on the unseen **2023–2026 Test Set**:

- **Root Mean Squared Error (RMSE):** Measures standard deviation of prediction residuals.
- **Mean Absolute Error (MAE):** Average error magnitude in runs/boundaries per over.
- **Coefficient of Determination (R^2 R2):** Proportion of variance explained by the model.

Compile these into a benchmark table:

```
csv
Target,Model,RMSE,MAE,R2
economy_rate,XGBoost,X.XX,X.XX,X.XX
economy_rate,Random Forest,X.XX,X.XX,X.XX
economy_rate,Linear Regression,X.XX,X.XX,X.XX
boundary_frequency,XGBoost,X.XX,X.XX,X.XX
```

🔹 STEP 7: Environmental & Pitch Impact Analysis (Feature Importance)

To demonstrate the spatio-temporal contribution:

- Extract **Gini / Gain Feature Importances** and **SHAP (SHapley Additive exPlanations)** values from the tree-based models.
- Quantify specific domain findings, such as:
    - *Does high humidity/dew significantly increase bowler economy rate in 2nd innings?*
    - *Does pitch degradation assist spin bowlers while hurting fast bowlers in middle overs?*

🔹 STEP 8: Visualizations & Deliverables

Generate and save high-resolution plots:

1. `outputs/model_comparison_rmse.png`: Bar chart comparing RMSE and R^2 R2 across all models.
2. `outputs/feature_importance_economy.png`: Top 15 most important features for Economy Rate.
3. `outputs/feature_importance_boundaries.png`: Top 15 features for Boundary Frequency.
4. `outputs/weather_vs_economy_actual_predicted.png`: Scatter/trend analysis of temperature/humidity vs. bowling performance.
5. `outputs/residual_distribution.png`: Error residual distribution plot.

📋 3. Quick Checklist for Project Review / Presentation

- Ingested the final fused dataset from teammate (PySpark module).
- Constructed non-leaking feature vectors (Cricket + Weather + Pitch Wear).
- Segmented train/test sets temporally (Train: 2008–2022, Test: 2023–2026).
- Trained 3 models (Linear Regression, Random Forest, XGBoost) per target.
- Generated `model_comparison.csv` with RMSE, MAE, and R^2 R2.
- Generated 4+ high-res visualization charts in `outputs/`.
- Documented the written Results & Discussion section explaining weather/pitch wear effects.